

Desmos Mastery - Day 5

10 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Regression Part 1 | 10 questions

[Lesson video: Regression Part 1](#)

1. A linear function passes through (2, 11) and (6, 23). What is its y-intercept?

Answer: _____

2. An exponential model $f(x) = ab^x$, with $a > 0$ and $b > 0$, satisfies $f(1) = 12$ and $f(3) = 108$. What is b ?

Answer: _____

3. A quadratic has vertex (3, -5) and passes through (5, 7). What is its leading coefficient?

Answer: _____

4. The model below satisfies $f(0) = 7$, $f(1) = 11$, and $f(2) = 19$. What is c ?

$$f(x) = ab^x + c, \quad a > 0, \quad b > 1$$

Answer: _____

5. Which dataset is sufficient to determine a unique quadratic $f(x) = ax^2 + bx + c$?

A) $f(1) = 4$ only

B) $f(1) = 4$ and $f(2) = 9$ only

C) $f(1) = 4$, $f(2) = 9$, and $f(3) = 16$

D) Three repeated copies of $f(1) = 4$

6. A circle passes through (1, 2), (5, 2), and (3, 4). What is the y-coordinate of its center?

Answer: _____

7. The function below satisfies $g(1) = 8$ and $g(4) = 2$. What is $g(2) + g(3)$?

$$g(x) = a(x - 2.5)^3 + b(x - 2.5) + c$$

Answer: _____

8. A line is modeled by $y = mx + b$ and passes through $(-2, 9)$ and $(4, -3)$. At what x-coordinate does it cross the x-axis?

Answer: _____

9. A numerical fit returns $a = 2$ and $b = 5$ for $y = ax^2 + b$. Which check would expose an incorrect fit to the points (1, 7) and (3, 25)?

- A) Verify the fit only at $x = 0$.
- B) Check the original equations at both given inputs.
- C) Accept the fit because both parameters are integers.
- D) Remove the point with the larger input.

10. A quadratic $f(x) = ax^2 + c$ has $f(-2) = 17$ and $f(3) = 42$. What is c ?

Answer: _____

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: 5 2: 3 3: 3 4: 3 5: C 6: 2 7: 10 8: 2.5 9: B 10: -3

Why each answer works

1. Answer: 5. The slope is $(23 - 11)/(6 - 2) = 3$. In $y = 3x + b$, the point $(2, 11)$ gives $b = 5$.

2. Answer: 3. Divide the values: $b^2 = 108/12 = 9$. Since $b > 0$, $b = 3$. Then $a = 4$; verify both original points.

3. Answer: 3. Use $f(x) = a(x - 3)^2 - 5$. Substituting $(5, 7)$ gives $7 = 4a - 5$, so $a = 3$. The vertex is a constraint, not just another ordinary point.

4. Answer: 3. Successive differences are 4 and 8, so $b = 2$. Then $a(b - 1) = 4$ gives $a = 4$, and $a + c = 7$ gives $c = 3$. All three values check.

5. Answer: C. Three distinct inputs impose three independent constraints on a quadratic. Repeated copies of a single observation do not add information.

6. Answer: 2. The first two points force the center onto $x = 3$. Equal distances from $(1, 2)$ and $(3, 4)$ give $4 + (k - 2)^2 = (k - 4)^2$, so $k = 2$.

7. Answer: 10. Let $z = x - 2.5$. Inputs 1 and 4 have opposite z -values, so their odd-power terms cancel; $g(1) + g(4) = 2c = 10$. Inputs 2 and 3 also have opposite z -values, so their sum is the same.

8. Answer: 2.5. The slope is $-12/6 = -2$. Using $(4, -3)$ gives $b = 5$. Set $-2x + 5 = 0$ to obtain $x = 2.5$.

9. Answer: B. At $x = 1$ the model gives 7, but at $x = 3$ it gives 23 rather than 25. A plausible numerical output must still satisfy the original constraints.

10. Answer: -3. The equations are $4a + c = 17$ and $9a + c = 42$. Subtraction yields $5a = 25$, so $a = 5$ and $c = -3$. The symmetric model has no linear term.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.